Kalshi Market Making

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Investment Universe

Kalshi KXINXU

Threshold contract

Pays \$1 if SPX closes **above** a specific level at 4:00 PM ET; \$0 otherwise.

YES $\equiv$ binary call on SPX.
NO $\equiv$ binary put on SPX.

Kalshi KXINX

Range contract

Pays \$1 if SPX closes **within a 25-point bracket** at 4:00 PM ET; \$0 otherwise.

YES $\equiv$ binary call spread on SPX.
NO $\equiv$ the complement (binary strangle).

SPY ETF

Delta hedge instrument

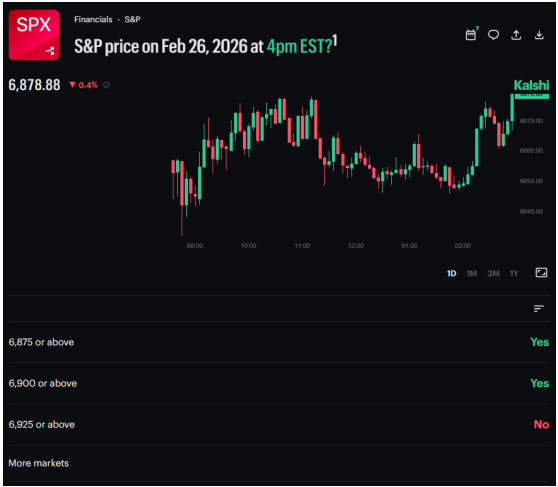
Tracks the S&P 500; $\partial \mathrm{SPY}/\partial \mathrm{SPX} \approx 0.1$.

We trade SPY shares to offset directional exposure from our Kalshi book.
Hedge size: $-10 \times \Delta_{\text{book}}$ shares.

Key Feature

All contracts settle at **4:00 PM ET** against the official SPX closing print. We operate exclusively in contracts with **0–1 day to expiration**, where the fair value equals the lognormal probability of SPX finishing above a given strike.

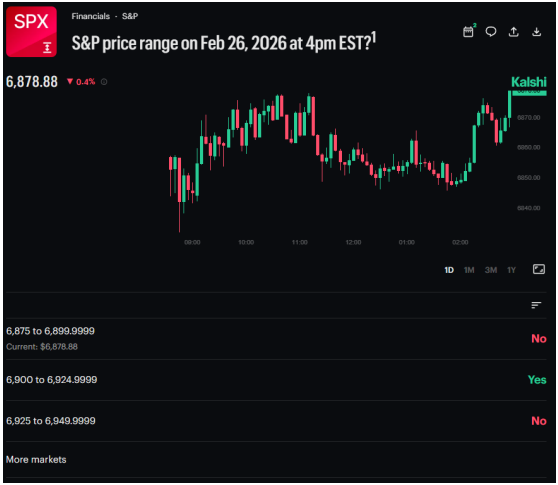
KXINXU — Threshold ("Above / Below") Contracts



KXINXU market for Feb 26, 2026 (resolved)

- Format: `KXINXU-26FEB26H1600-T6875` (threshold = 6875, expiry = Feb 26 4:00 PM)
- **YES** pays \$1 if SPX > threshold; \$0 otherwise
- **NO** pays \$1 if SPX ≤ threshold; \$0 otherwise
- YES ≡ binary call; NO ≡ binary put
- Prices reflect the market's probability that SPX will exceed the threshold
- We simultaneously quote bids and offers across many strike levels

KXINX — Range ("Bracket") Contracts



KXINX market for Feb 26, 2026 (resolved)

- Format: `KXINX-26FEB26H1600-B6987` (midpoint = 6987, bucket width = 25 pts)
- **YES** pays \$1 if SPX $\in [\text{mid} - 12.5,\ \text{mid} + 12.5)$
- Replicated as:
  - Long binary call at lower strike K_L
  - Short binary call at upper strike $K_U = K_L + 25$
- **NO** is the complement: a binary strangle
- Range contracts concentrate near the current SPX level and converge sharply to \$0 or \$1 through the day

Strategy Overview

What we do

We run an **automated market-making** strategy on Kalshi SPX event contracts: posting simultaneous bids and offers around a model-derived fair value, capturing the bid-ask spread as profit.

Core loop (every second):

1. Compute fair value using live SPX and VIX
2. Post bid and ask around the fair value
3. Detect fills from market order flow
4. Compute book delta; send SPY hedge order
5. Settle expired contracts at 4 PM

What we are not doing:

- Directional betting on SPX
- Predicting news events
- Using machine learning signals

Our profit source:

- Bid-ask spread on each filled round-trip
- Delta hedge isolates spread P&L from SPX moves

Competitive Edge

Retail-Dominated Counterparty Flow

- Kalshi weekly volume hit a record **$3.7B** in January 2026
- Estimated **90%+** of volume from sports & entertainment betting — counterparties lack quantitative pricing models
- Retail traders react to news and emotion rather than BSM fair values, creating persistent mispricings
- Prediction markets are **new territory**: far fewer systematic market makers than in equity or options markets

Structural Pricing Advantage

- We use **live SPX + VIX** to compute risk-neutral probabilities in real time — most counterparties do not
- At 0–1 DTE, contract prices equal the lognormal probability of SPX exceeding a strike; retail traders frequently deviate
- Our mean absolute pricing error vs. VWAP is **7.1 cents**, confirming our model is a reliable anchor
- As Kalshi matures, designated **market-maker programs** may provide additional structural compensation

The Model: Binary Option Pricing

A Kalshi YES contract is a **binary call** on SPX. Under risk-neutral pricing, its fair value at time t is:

$$V_t = e^{-\int_t^T r(u)\,du}\,\mathbb{Q}_t\big(S_T > K\big)$$

- $e^{-\int_t^T r(u)\,du}$ — discount factor (negligible at 0–1 DTE)
- $\mathbb{Q}_t(S_T > K)$ — risk-neutral probability of SPX finishing above strike K

Key simplification at 0–1 DTE

The discount factor $\to 1$ as $T - t \to 0$, so

$$V_t \;\approx\; \mathbb{Q}_t\big(S_T > K\big)$$

The contract price is simply the **market-implied probability** of SPX finishing above the strike — a number between 0 and 1.

The Model: Contract Replication

KXINXU (Threshold)

$$\text{YES} = \text{Binary call at strike } K$$
$$\text{NO} = 1 - \text{YES}$$

Example: "SPX above 6875" YES
$\Rightarrow$ Binary call, strike $K = 6875$

One formula prices all threshold contracts.

KXINX (Range Bracket)

$$\begin{aligned}\text{YES} &= \mathbb{Q}(K_L < S_T \leq K_U) \\ &= P(S_T > K_L) - P(S_T > K_U)\end{aligned}$$

where $K_U = K_L + 25$ pts

Replication: Long binary call at K_L, short binary call at K_U

Implication

Every Kalshi SPX contract can be priced using **one formula**: the risk-neutral probability $\mathbb{Q}(S_T > K)$ under a lognormal SPX model.

The Model: Lognormal CDF (Black-Scholes)

Following the Black-Scholes-Merton framework, we model the required probability as:

$$\mathbb{Q}_t(S_T > K) = \Phi(d_2) \qquad \text{where} \qquad d_2 = \frac{\ln(F/K) - \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}$$

Inputs (updated every second):

- $F \approx S_t$ — current SPX level (no dividend adjustment at 0–1 DTE)
- K — contract strike (threshold or bucket edge)
- $T - t$ — time to 4:00 PM expiry in years
- $\sigma = \text{VIX}/100$ — implied vol from CBOE VIX

Why VIX for volatility?

- VIX is the market's real-time estimate of 30-day SPX implied vol
- At 0–1 DTE we assume flat term structure across all strikes
- Simple, observable, and updated every second
- Automatically widens our spreads when market uncertainty rises

The Model: Spread Design & Fees

Spread components:

- **Base spread** (2 cents) — minimum compensation
- **VIX adjustment** ($+0.1\times\text{VIX}/100$) — wider when volatile
- **Inventory adjustment** ($+0.025\%$ per contract) — wider as book grows
- **After-hours widening** ($+10$ ticks) — outside 9:30–16:00 ET

Inventory skew:

- Long inventory $\Rightarrow$ lower mid to attract sellers
- Prevents unbounded one-sided accumulation
- Bid/ask sizes also shrink as inventory grows

Kalshi maker fee:

$$\text{fee} = \left\lceil 0.0175 \times C \times P(1-P) \right\rceil_{\text{cent}}$$

- Highest at-the-money ($P \approx 0.50$); near zero deep ITM/OTM
- Fee is built into every quote so net P&L after fees is positive

Fee impact

Maker fees consumed **\$684.47 (41.6%)** of gross spread earnings in our backtest. Fee incorporation is non-negotiable.

The Model: Pricing Flow

Live SPX level & CBOE VIX (updated every second)

↓ Lognormal CDF

$$d_2 = \frac{\ln(S_t/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}} \quad \Rightarrow \quad \Phi(d_2) = \mathbb{Q}(S_T > K)$$

↓ Risk-neutral pricing

Fair value for each binary call / put

↓ Call spread replication

Fair value for each KXINX(U) contract via replication

↓ Spread model + fee adjustment

Bid & ask = fair value ± dynamic half-spread (VIX + inventory + fees)

The Data: Sources

Series	Source	Freq.
SPX index	Bloomberg	1-second
VIX index	Bloomberg	1-second
SPY OHLCV	Databento	1-second
Kalshi KXINXU trades	Kalshi API	Trade event
Kalshi KXINX trades	Kalshi API	Trade event

Coverage & alignment:

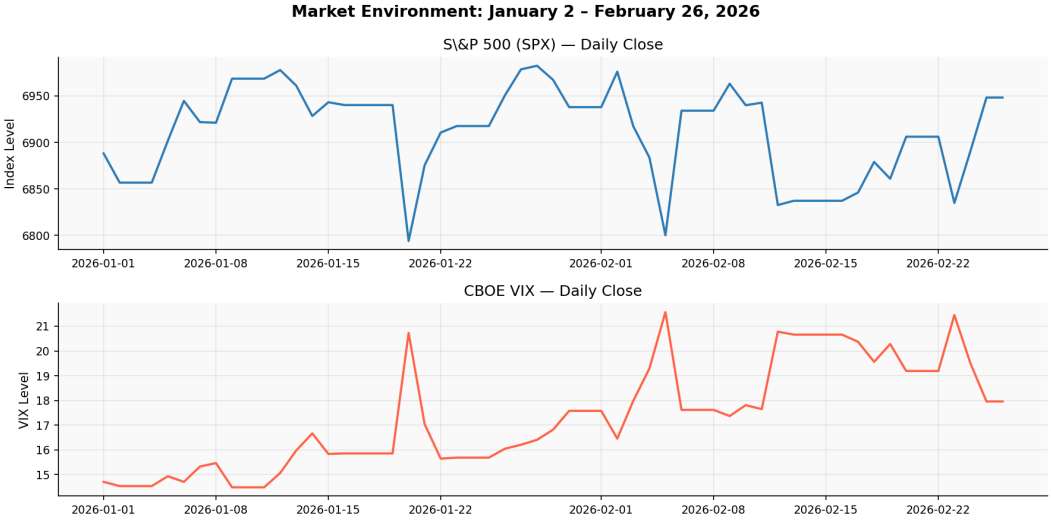
- **Period:** Jan 2 – Feb 26, 2026 (38 trading days)
- SPX/VIX: extended hours; SPY: RTH only (9:30–16:00 ET)
- Kalshi events aggregated to 1-second bars
- All timestamps UTC-normalized, 1-second grid

Metric	Value
SPX/VIX records	767,537
Total Kalshi trades	56,742
in RTH	87.9%
after-hours	12.1%
SPX range	6,797 – 6,979
VIX range	14.5 – 21.5
VIX mean	17.6

Data quality:

- SPY forward-filled outside RTH
- Rows with missing macro data dropped

The Data: Market Environment



SPX trended upward +0.74% over the period. Two clusters of elevated VIX (mid-January, early February)

Market Making Logic

Every second, for each active contract:

1. Retrieve latest fair value from Pricer
2. Compute spread: base + VIX adj. + inventory adj.
3. Apply inventory skew to shift mid
4. Round bid **down**, ask **up** to nearest cent
5. Clamp to $[0, 1]$ (binary contract bounds)
6. Incorporate Kalshi maker fee into each leg
7. Reduce bid/ask sizes as inventory grows
8. If cash $< 5\%$ of capital: suspend bids

Spread parameters:

Parameter	Value
Base spread	2 cents
Tick size	1 cent
VIX slope	0.001 / VIX pt
Inventory slope	0.00025 / contract
Inventory skew	0.0005 / contract
After-hours add	+10 ticks

Example: VIX $= 20$, 40 YES contracts held

$\Rightarrow$ spread $= 2 + 2 + 1 =$ **5 cents**
$\Rightarrow$ mid skew $= -2$ cents (sell pressure)

Simulation Logic: Key Assumptions

Fill Assumption (Last-in-Queue)

We fill only when a market order is priced **at least one tick better** than our posted quote:

$$\text{Bid fill: } \text{take_bid} \leq \text{my_bid} - \$0.01$$
$$\text{Ask fill: } \text{take_ask} \geq \text{my_ask} + \$0.01$$

Conservative: assumes we are **last in the queue**.

Fill Size Cap

Fill quantity is capped at the observed market trade size per second.

Execution Delay

SPY hedge orders execute with a **1-second delay** to reflect realistic latency. Kalshi fills apply immediately.

Settlement

Contracts expiring at 4:00 PM ET settle at the next day boundary. Payouts:

- KXINXU YES: \$1 if SPX $>$ threshold, else \$0
- KXINX YES: \$1 if SPX falls in range, else \$0
- NO contracts: 1− YES payout

Delta Hedging Logic

Step 1 — Contract delta

$$\Delta_{\text{YES ABOVE}}(K) = \frac{\varphi(d_2)}{S_t\,\sigma\,\sqrt{T-t}}$$

- $\varphi(\cdot)$ = standard normal PDF
- Units: contract value change per \$1 SPX move
- NO contracts: $\Delta_{\text{NO}} = -\Delta_{\text{YES}}$
- Range contracts: difference of two call deltas

Step 2 — Book delta & SPY hedge

$$\Delta_{\text{book}} = \sum_i q_i\,\Delta_i$$

Since $\partial\text{SPY}/\partial\text{SPX} \approx 0.1$:

$$\text{SPY target} = -10 \times \Delta_{\text{book}} \text{ shares}$$

- Negative book delta $\Rightarrow$ buy SPY
- Positive book delta $\Rightarrow$ sell SPY short
- Hedge submitted every second with 1-second delay
- Reduces tail risk by $\approx 30\%$ (VaR 95%)

Leverage & Shorting Logic

Kalshi Positions

- **Long YES/NO**: pay face value at fill; settle at \$0 or \$1
- **Short YES/NO**: receive face value; owe payout at settlement
- Binary structure means maximum loss per contract is known in advance
- **Cash guard**: bids suspended when cash $< 5\%$ of initial capital

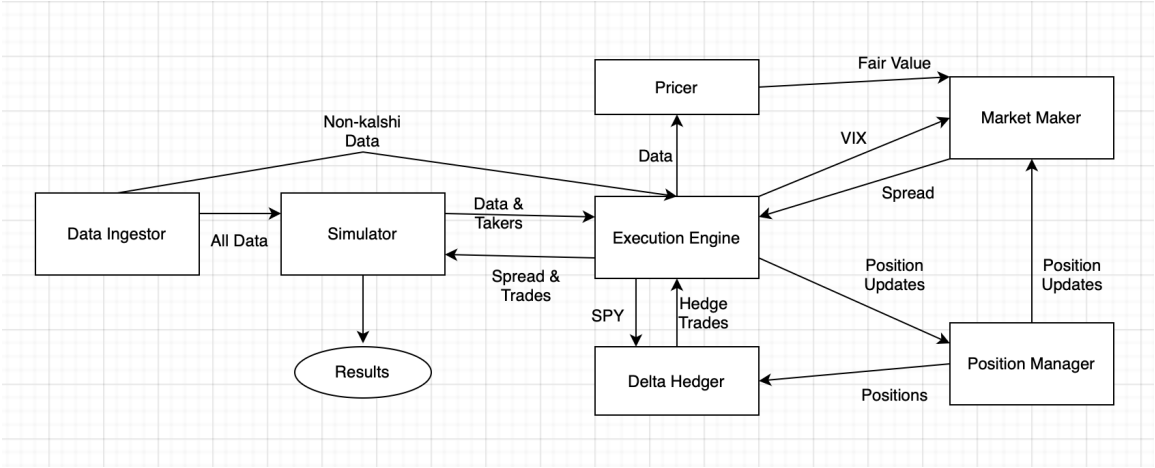
SPY Short-Selling Margin

- **Initial margin**: 50% of notional to open a short SPY position
- **Maintenance margin**: 30% of notional held at all times
- Hedge order skipped if available cash is insufficient
- Long SPY uses cash directly; no leverage applied

Leverage Policy

No leverage on Kalshi positions. Profitability comes from spread capture, not capital magnification. Margin rules are conservative proxies for live trading.

Program Architecture



Risk Management

Position Rules

- Bid size shrinks as inventory grows
- Bids suspended when cash $<$ \$500 (5% of \$10k)
- SPY shorts require 50% initial margin
- Maintenance margin: 30% of short SPY notional

Delta Hedging

- Book delta targeted to zero every second
- Hedge reduces VaR 95% by $\approx$30%: \$4.49 vs. \$6.44 (unhedged)
- 1-second delay leaves brief unhedged exposure during fast moves
- After hours: wider spreads compensate for unavailable SPY hedge

Spread Widening

- VIX spike $\Rightarrow$ automatic spread widening
- Growing inventory $\Rightarrow$ wider spread + mid skew
- After hours: +10 ticks to compensate for unhedged exposure
- Near settlement: probabilities converge, spreads widen naturally

Backtesting Methodology & Assumptions

What is included:

- **Kalshi maker fees** ($\lceil 0.0175 \times C \times P(1-P) \rceil$)
- **1-second execution delay** on SPY hedge trades
- **Last-in-queue fill assumption** (conservative)
- **Margin requirements** for SPY shorts (50% initial, 30% maintenance)
- **Cash constraints** (no bids if cash $< 5\%$ of capital)
- Mark-to-market using live Pricer fair values

Period: Jan 2 – Feb 26, 2026 (38 trading days)
Capital: $10,000 initial

Simplifications / omissions:

- No SPY transaction costs (commissions, spread)
- Flat vol term structure (VIX used for all strikes)
- No vol smile / skew (BSM lognormal assumption)
- No Kalshi exchange outages or connectivity issues
- Adverse selection not explicitly penalized
- Two-month window may not capture all market regimes
- No train / test split — all data used in-sample

Disclosure

Past simulation performance does not guarantee future results. Live trading adds costs and operational risks not captured here.

Results: Performance Summary

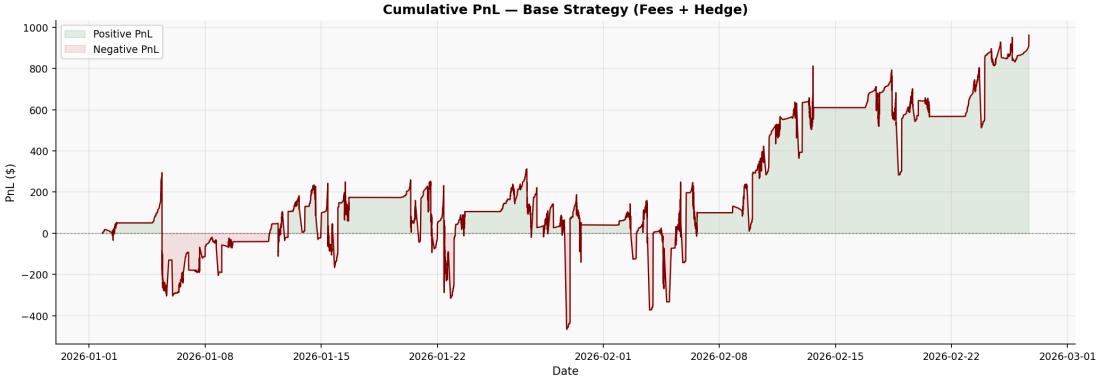
Metric	Base (fees+hedge)	No Fees (hedge on)	No Hedge (fees on)
Net PnL ($)	**$961.20**	$1,645.67	$1,425.84
Period return	**9.61%**	16.46%	14.26%
Sharpe ratio	**6.14**	10.00	7.22
Sortino ratio	**5.71**	9.29	6.56
Max DD ($)	**$778.56**	$772.02	$1,090.67
Max DD (%)	**7.55%**	7.55%	10.56%
Win days	**25 / 38**	—	—
Day win rate	**65.8%**	—	—

Sharpe annualized from 1-second returns; $r_f = 4.25\%$ *p.a.*

Key Takeaways

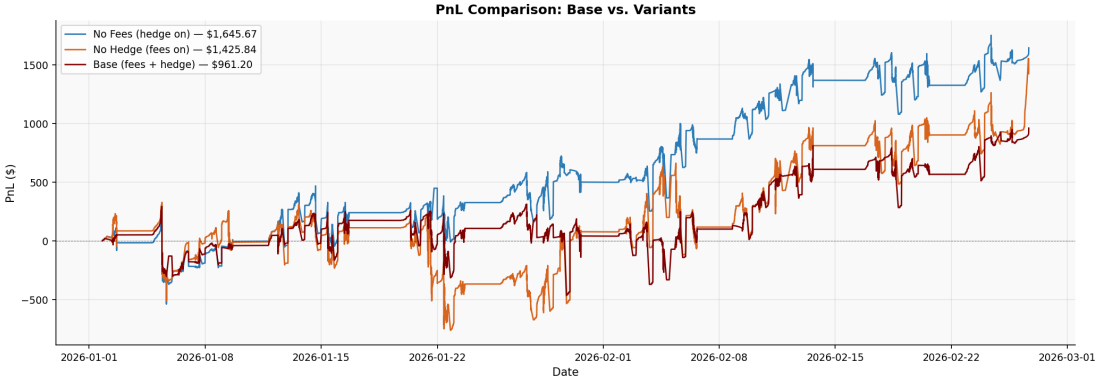
- **Base** earned $961.20 (9.61%) net of all costs over 38 days
- **Fees** consumed $684.47 — 41.6% of gross earnings
- **Hedging** cost $464.64 in PnL but cut Max DD from 10.56% to 7.55%
- Sharpe of **6.14** is well above typical benchmarks
- Profitable on **65.8%** of trading days

Results: PnL Over Time



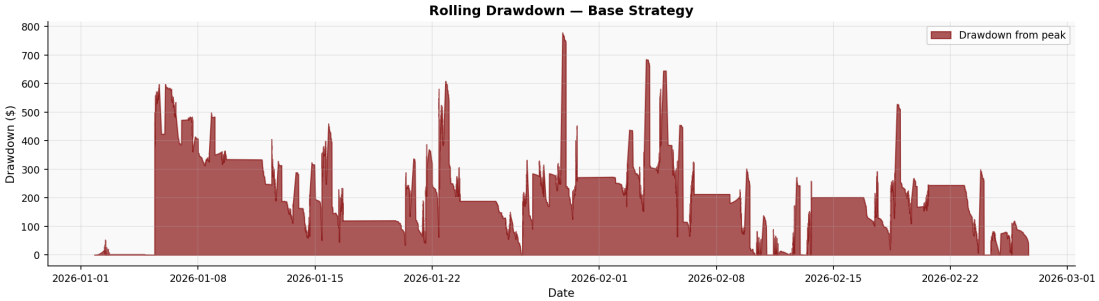
PnL grew steadily with two notable drawdown episodes coinciding with elevated VIX. Largest drawdown: $778.56 (7.55%), mid-January.

Results: Strategy Variants Comparison



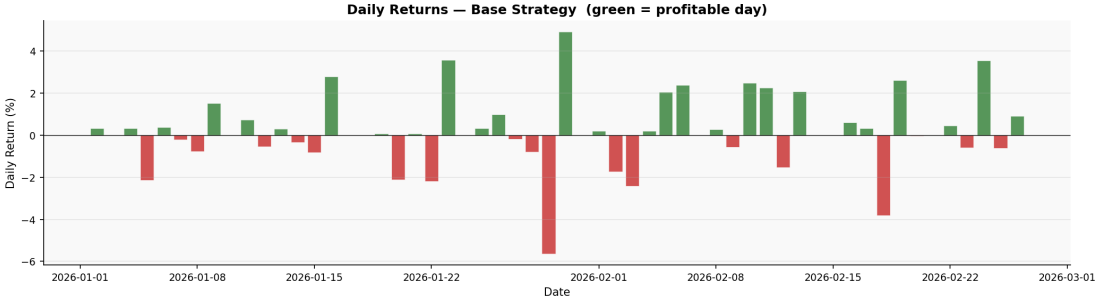
No-fees confirms the gross edge ($1,645.67). **No-hedge** is profitable but drawdown grows to 10.56% and SPX correlation rises, confirming the hedge's risk-reduction value.

Results: Drawdown Analysis



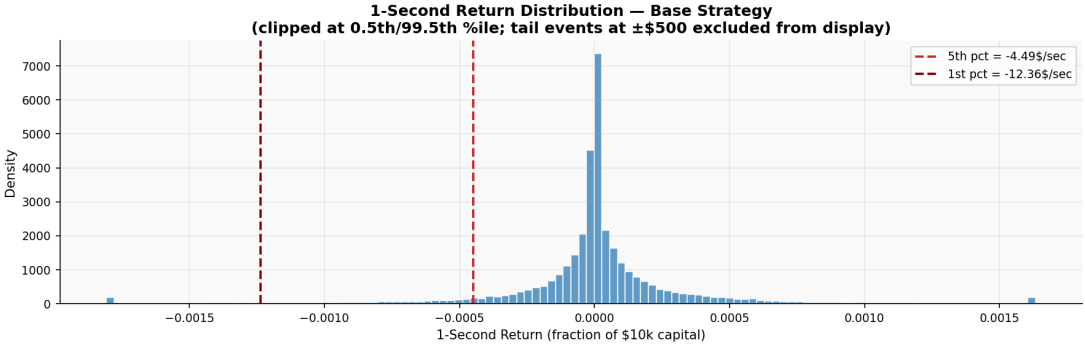
Drawdowns are short-lived, primarily driven by fast SPX moves before the delta hedge adjusts. The strategy recovers quickly as spreads are recaptured on subsequent fills.

Results: Daily Returns



25 of 38 days (65.8%) profitable. Negative days cluster around mid-January and early-February VIX spikes, consistent with brief unhedged directional exposure.

Results: Return Distribution



Returns tightly clustered near zero with heavy tails. Tail events ($\pm\$500$) are settlement moments when large inventories resolve at once. 5th percentile: $-\$4.49/\textbf{sec}$; 1st percentile: $-\$12.36/\textbf{sec}$.

Results: Risk Metrics (VaR & CVaR)

Scenario	VaR 95% ($/s)	CVaR 95% ($/s)	VaR 99% ($/s)	CVaR 99% ($/s)
Base (fees + hedge)	−4.49	−12.75	−12.36	−35.46
No Fees (hedge on)	−4.65	−13.63	−13.45	−38.60
No Hedge (fees on)	−6.44	−18.11	−18.13	−49.69

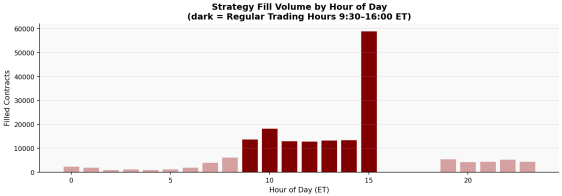
Interpretation

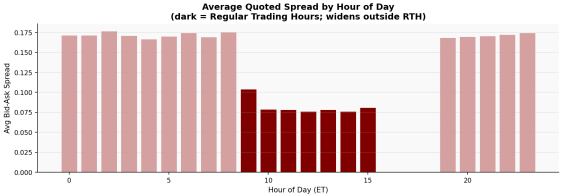
- **VaR 95%**: worst 5% of seconds lose at least $4.49
- **CVaR 95%**: average loss in that worst 5% is $12.75
- Delta hedge cuts worst-case loss by **30%** ($6.44 → $4.49)
- Tail events driven by settlement of large inventories at 4 PM

SPX Correlation

- 1-second: −0.19 (base), −0.26 (no hedge)
- Daily: +0.14 (base), +0.32 (no hedge)
- Low correlation confirms alpha is **structural** spread capture, not directional SPX exposure

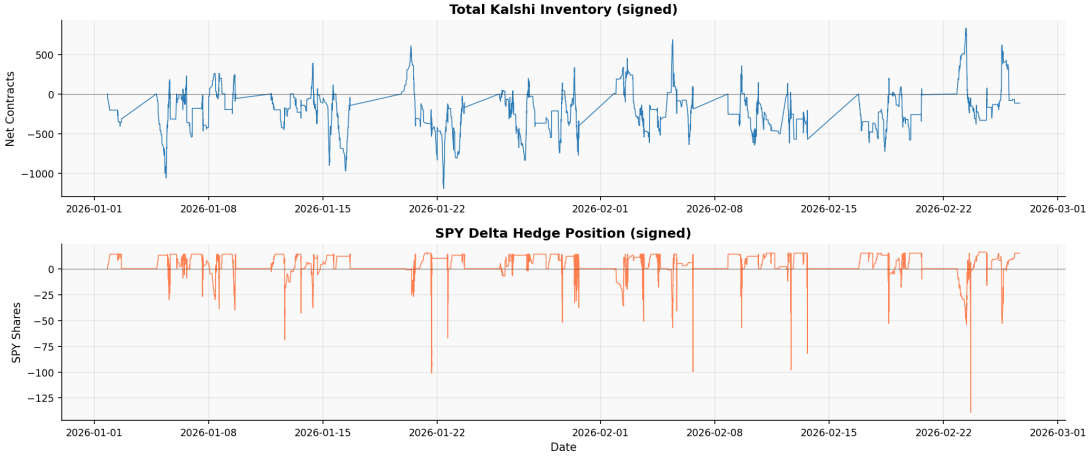
Results: Intraday Activity





Left: Fills peak at open (9:30 ET) and close (15:00–16:00 ET). **Right:** Spreads widen outside RTH (+10 ticks) and at open/close when inventory risk is highest. Total fills: **4,628** over 38 days.

Results: Inventory & Hedge Position



Top: Net Kalshi inventory fluctuates around zero — the strategy stays roughly balanced. **Bottom:** SPY position mirrors inventory delta; accumulations are offset by hedge trades.

Results: Fee & Hedge Attribution

Scenario	PnL ($)	vs. Gross	Note
Gross (no fees, no hedge)	$1,645.67	baseline	Spread captured before costs
No Hedge (fees on)	$1,425.84	−$219.83	Fee drag only
No Fees (hedge on)	$1,645.67	−$0	(same as gross; hedge cost via PnL)
Base (fees + hedge)	**$961.20**	−$684.47	**Net after all costs**

Fee Impact

Maker fees consumed **$684.47 (41.6%)** of gross. Most of the fee drag occurs on near-the-money fills where $P \approx 0.50$. Fee-efficient quoting (e.g., wider spreads at-the-money) is the primary lever for improvement.

Hedge Impact

The hedge cost **$464.64** in realized PnL (SPY losses when SPX trends) but cut Max DD from **10.56% to 7.55%** and VaR 95% by 30%.

Key Findings

1. **Profitable:** \$961.20 net (9.61% on \$10k) over 38 days, Sharpe 6.14
2. **Clear gross edge:** \$1,645.67 before fees confirms persistent mispricings vs. BSM
3. **Fees are the primary drag:** 41.6% of gross — fee-efficient quoting is critical for live deployment
4. **Delta hedge reduces risk:** Max DD 7.55% (hedged) vs. 10.56% (unhedged); VaR 95% improves $\approx$30%
5. **Alpha is structural, not directional:** SPX correlation -0.19 (1s), $+0.14$ (daily) — spread capture, not market beta
6. **High-VIX sessions most profitable:** elevated uncertainty widens spreads while fill rate stays stable
7. **Consistent edge:** profitable 25 of 38 days (65.8%); rolling Sharpe broadly positive throughout
8. **Adverse selection present:** 79% of mid-price moves 1s post-fill are adverse — counterparty monitoring needed